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An Unexpected Thermal Event in the Khaleej Al Bahrain Basin, Arabian Plate

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Abstract

A recent study investigating Upper Jurassic source rocks in Southwest Bahrain (Khaleej al Bahrain basin) reveals that subsurface temperatures were once significantly higher – by around 20-40°C – than they are today. This discrepancy suggests a previously undocumented "past thermal event," which may hold major implications for understanding the regional petroleum system and could influence future hydrocarbon exploration efforts. To uncover evidence of this thermal anomaly, the research team employed a suite of advanced geochemical techniques. These included organic petrography (bitumen reflectance), thermal Rock-Eval® (Shale Play™ method), bulk kinetic analysis using extracted kerogen, fluid inclusion studies, and oil and biomarker analysis of both rock extracts and crude oil samples. Additionally, a comprehensive 3D basin model (TemisFlow®) was created and calibrated using pressure and temperature data from wells to integrate these findings into the geological context. Key results indicate that thermal maturity levels exceed 0.8 to 0.9% VRoeq (peak oil window) at a burial depth of 7,500 feet, despite current temperatures at this depth being only 95–100°C – insufficient to cause such maturity. Fluid inclusion data show past temperatures of 135°C at the same depth. At 10,000 feet, current maturity levels between 1.15% and 1.3% VRoeq cannot be explained by the present thermal gradient (~30°C/km). The basin model, coupled with a lithospheric model, makes it possible to test hypotheses about the nature of "past thermal event" that could explain the observed anomalies. One possible explanation is the Arabian Plate's movement over the Afar Hotspot, which may have led to localized thermal anomalies. An alternative explanation could be a significant erosion of over 4,000 feet of the Dammam Formation during the Oligocene, which is considered unlikely and less supported by data. This proposed thermal event has not been previously documented and warrants further investigation. If validated, it could reshape current geological models of Bahrain and surrounding areas, challenging the notion of thermal stability in similar sedimentary basins and offering new perspectives for exploring unconventional hydrocarbon plays.

Introduction

A project conducted in 2020 between the national operator, Tatweer Petroleum (now BAPCO), and IFPEN, in collaboration with Beicip-Franlab & IFPMEC, aimed to enhance the understanding of the conventional and unconventional Jurassic plays – specifically the Hanifa and Tuwaiq Mountain formations – at the country scale. An additional objective was to consolidate Bahrain's national geochemical database through new geochemical analyses. The study area encompasses the entire territory of Bahrain (onshore & offshore). Geologically, Bahrain is located on the northeast of the Arabian Plate, at the edge of the foreland basin system associated with the Zagros orogeny. The island is structurally dominated by the Awali Dome, a broad anticline situated between the Qatar Arch to the east and the Khaleej al Bahrain Basin to the west, which extends into Saudi Arabia as the Jafurah Basin. The oldest penetrated sediments date back to at least the Lower Paleozoic, and the sedimentary record is relatively continuous, except for significant erosional episodes during the Carboniferous, Triassic, and Neogene. The Jurassic interval of interest is characterized by extensive carbonate deposition, with source rocks such as the Tuwaiq Mountain Formation (Tuwaiq-SR) deposited in intra-platform depressions within the carbonate system.

The work was divided into two phases: (1) analytical experimental work performed by IFPEN laboratories (organic geochemistry, source rock and oil characterization); (2) unconventional play assessment performed by Beicip-Franlab (based on basin and petroleum system modelling solutions). Beyond the expected play analysis results, the study highlighted thermal maturity anomalies. This manuscript focuses on this element of the study.

Thermal maturity is the degree of thermal alteration of sedimentary organic matter within a rock, indicating its capability (past and future) to generate hydrocarbons. It characterizes "thermal stress" and depends on the evolution of temperature during geological times – the transformation of organic matter being a kinetic process, the notion of duration of exposure is essential. It often happens that the current temperature is less than the maximum temperature and therefore that maturity is greater than what the current temperature field could suggest. It is a key factor in petroleum system analysis.

Several techniques are used to measure the thermal maturity (maturity indicators or markers): maceral reflectance measured in the scope of organic petrography studies (usually vitrinite reflectance VRo); interpretation of source rock pyrolysis (such as Rock-Eval®), sometimes including the analysis of kerogen kinetic schemes; observations of the colors of different kinds of organic matter (e.g., SCI or Spore Color Index, TAI or Thermal Alteration Index); Raman spectroscopy (especially for highly mature samples); biomarker analysis (certain molecular ratios, as determined by GC-MS analysis of produced oils and fluids extracted from source rocks, can serve as indicators of the thermal maturity or thermal stress experienced by the fluids, either within the host rock or along their migration pathway); gas composition and isotopic signature (including mud gas). Other techniques make it possible to estimate this thermal maturity indirectly, without analyzing organic matter. Let us quote for example: the analysis of fluid inclusions (in particular the measurement of the homogenization temperature in fluid inclusions), sometimes coupled with a dating of the mineral phase containing fluid inclusions; all thermochronometry studies which allow an interpretation of time elapsed after the closure of a temperature-dependent geochemical system (for example Apatite Fission Tracks Analysis AFTA, U-Th/He systems on zircon and apatite, etc.), and therefore the recording of the cooling and the exhumation of the rocks. Several of these techniques have been implemented as part of this study, others are planned for future work. It is always interesting to combine different techniques and data from different laboratories to improve the reliability of the analysis. Indeed, the measurement of maturity is sometimes subjective and always sensitive to experimental biases. For the sake of simplicity, we will use vitrinite reflectance as the reference maturity scale, with other maturity indicators converted to equivalent vitrinite reflectance (VRoeq).

The integration and interpretation of the different maturity indicators are carried out using numerical models capable of simulating the thermal history of sedimentary basins (basin models), and thus calculating the thermal stress experienced by rocks. The experimental results serve both as a basis for developing conceptual models to explain the observed maturity level and as calibration data to validate these conceptual models using numerical simulations. Finally, basin modelling is used here as a tool to test geological scenarios, with the aim of predicting maturity levels away from the wells and assessing the petroleum potential of the Jurassic interval at country scale.

After briefly describing the study methodology, we will present the results of the experimental program used to assess and map maturity levels (noting that only partial results can be shared in this publication). We will then propose and test geological scenarios to explain the observed maturity levels. Finally, we will evaluate the potential impact of the identified thermal maturity anomalies on our understanding of the regional petroleum systems, and beyond.

Methodology & Tools

Recovery of Existing Data

The initial dataset provided for this study includes a comprehensive suite of subsurface and geochemical information essential for petroleum system analysis. It comprises well reports from approximately 20 wells distributed across the country, which include well markers as well as temperature and pressure data obtained from well tests. The analysis of temperature data is particularly critical in this study. In addition, 19 interpreted seismic horizons in depth (from the surface to the basement) are available at the country scale (although interpretations of the deeper horizons remain uncertain). Geochemical reports from previous studies are also included, particularly Rock-Eval® pyrolysis results and thermal maturity data. At the start of the study, the geochemical database already contained information from more than 20 wells, including over 400 Rock-Eval measurements and around 100 maturity indicators (VRoeq and SCI). The database also comprised more than 20 GC-MS analyses, primarily on oils from the Awali Field (Cretaceous and Jurassic reservoirs), along with a few analyses on source rock extracts from the Tuwaiq-SR.

The precise list of data cannot be disclosed here. For the purposes of this presentation, we will focus on five wells – A, B, C, D, and E – that form a south–north transect across the Awali Dome, the main structural feature of the country (Figure 1). Well A and Well E are located away from the anticline, with Well A situated in the Khaleej Al Bahrain basin and with Well E situated in the far north. Well B lies on the southwestern flank of the anticline, while Well C and Well D are positioned on its crest, where the Jurassic levels are shallowest (and the least mature). Well A, Well B and Well D contain key core samples selected for advanced geochemical analysis. Well C primarily contains data from deep Paleozoic intervals, which are particularly useful for thermal calibration of the model.

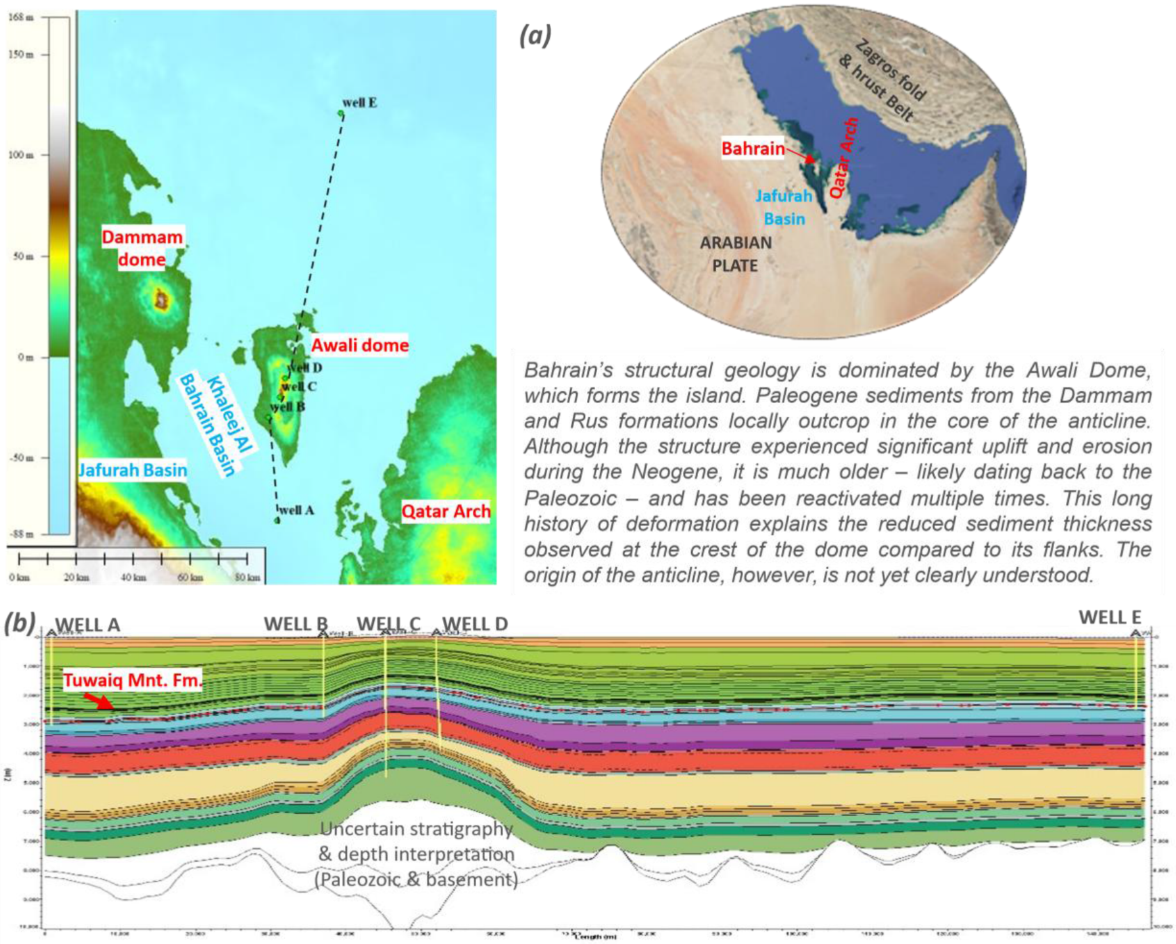


Figure 1—Study Area: (a) Topography map and selected wells location; (b) Cross section through the geological model.

Experimental Program

The primary objective of the experimental work carried out by IFPEN laboratories during Phase 1 of the project was the source rock characterization (including the thermal maturity level determination). First, Rock-Eval® 6 analysis were performed using the Shale Play™ method on 500 rock samples from 13 wells (cuttings and core samples, mostly from Tuwaiq Mountain Fm. & Hanifa Fm.); then bulk kinetic models on two selected core samples from Well B & Well D (organic-rich core samples from Tuwaiq Mountain Fm, showing high TOC and low OI) were obtained from the isolated kerogen fraction using the "Optkin" method. Additional geochemical analysis were performed as follow: organic petrography and palynofacies analysis, along with measurement of maceral reflectance over 50 samples (representative selection of cuttings and cores, bitumen reflectance BRo converted in VRoeq); oil extractions on 2 samples from Well B & Well D (selected organic-rich core samples from Tuwaiq-SR, high TOC and low OI), followed by Saturate/Aromatic/Resin separations, GC-MS & GC-IRMS analysis of the source rock extracts (classical biomarker analysis on Saturates and Aromatics, stable carbon isotopes $\delta^{13}C$). More details on the experimental procedures are available in the following published papers, for instance: Romero-Sarmiento et al. (2014; 2015; 2016; 2017) for the Rock-Eval®, Littke et al. (2012) for the organic petrography, Behar et al (2001) and Romero-Sarmiento et al. (2016) for the kinetic analysis, Romero-Sarmiento et al. (2017) for the molecular geochemistry.

In addition, 2 core samples from Well B displaying calcite veins were selected for fluid inclusion analysis (manufacturing of 100-120 μm thick sections, identification of different families of fluid inclusions, microthermometry, homogenization temperature).

Furthermore, the experimental program comprises the analyses of 11 oil samples from well tests in the Jurassic interval (10 different wells). API gravity, viscosity, and sulfur content were also measured. Finally, SAR separations were carried out on the 11 oil samples, followed by GCMS analysis for oil-source correlation based on Saturates and Aromatics biomarkers. This presentation is mainly focused on biomarkers associated to the maturity level, especially C29Ts/(C29Ts+C29H) & Ts/(Ts+Tm) – Terpanes m/z 191.

Basin Modelling (Physic-Based Numerical Modelling)

Basin modelling aids in understanding the formation and evolution processes of sedimentary basins. It simulates the geological, physical, and geochemical phenomena that drive basin evolution over geological time. This technique is a form of forward modelling, meaning that the simulation begins with the deposition of the first sediments in the basin and continues up to the present day. Basin modelling is used to determine the presence and quality of petroleum system components (Petroleum System Modelling), but it is also frequently applied for thermal modelling and pore pressure prediction. It is particularly useful at basin scale and in complex geological settings where well data are sparse or traditional geophysical approaches prove inadequate. In this study, a lithospheric model is fully coupled with the basin model to improve the accuracy of past thermal predictions by better accounting for the blanketing effect of sedimentation, disturbances from erosional events, and lithospheric-scale processes such as rifting (cf., Dubille et al., 2020). The basin modelling tool used in this study is TemisFlow® 1D & 3D (OpenFlow Suite).

The basin modelling workflow begins with the construction of the geological model – a coherent, dynamic representation of the subsurface from basement to surface – integrating geochemical, sedimentary, petrophysical, and geophysical data. Each model corresponds to a geological scenario (including past geological events) that is tested using numerical simulations. The initial simulations focus on pore pressure and thermal field modelling, followed by the geochemical modelling of source rocks, including the assessment of maturity levels. Hydrocarbon retention modelling is then performed (in source rocks to evaluate unconventional shale play potential), along with the modelling of hydrocarbon migration and accumulation (in this case following the "Darcy Flow" algorithm). The modelling results are compared against measured data, particularly temperature and maturity indicators. Calibration is an iterative, trial-and-error process: if the model does not align with observed data (assuming the data are reliable), the proposed geological model is considered incorrect and must be revised. When the data is uncertain (this is the case here), the calibration also includes a margin of uncertainty which must be analyzed. In some cases, the model can also be used to identify inconsistent data (a frequent case with thermal maturity data). This presentation focuses specifically on thermal modelling.

Experimental Results, Maturity Level Assessment

Rock-Eval® Data (Thermal Organic Matter Properties)

The relationships between depth and Rock-Eval® parameters – such as Tmax (the temperature in °C corresponding to the peak of hydrocarbon generation during the artificial pyrolysis), HI (Hydrogen Index, geochemical proxy used to characterize both kerogen origin and type, also associated to the residual hydrocarbon generation potential, expressed in mgHC/gTOC), PI (Production Index, expressed as a percentage, which indicates the proportion of free hydrocarbons present in the sample before pyrolysis relative to its generation potential, and thus provides a minimum estimate of the Transformation Ratio) – offer valuable insights into source rock maturity. Under increasing thermal stress in geological settings, an inverse relationship typically emerges between Tmax and HI, reflecting the progressive depletion of residual hydrocarbon potential and the concurrent increase in hydrocarbon generation by the source rock. However, this maturity trend becomes clearly discernible only when the samples possess a sufficiently high initial hydrocarbon potential (i.e., high-enough S2 peak, which correlate with high TOC and high HI), and when the kerogen is relatively homogeneous. Core samples also give more reliable results than cutting samples which are sometimes contaminated and altered. It is important to note that Rock-Eval® data offer relative maturity indicators that are strongly dependent on kerogen type.

In this study case, the data indicate that the Tuwaiq-SR kerogen varies between two endmembers: "Tuwaiq-Rich" and "Tuwaiq-Poor", probably due to change in the organic matter preservation, which introduces ambiguity into the interpretation in term of maturity level. Moreover, the hydrocarbon generation potential in the northern part of the country is too low to draw firm conclusions in this area. However, when focusing only on better quality core samples with high TOC and low Oxygen Index (e.g., samples from Well A, Well B & Well D between the Awali Dome and the Khaleej al Bahrain Basin with TOC >1%, OI <20 mg/g), a distinct maturity trend becomes clearly apparent, especially by plotting the HI vs. depth (Figure 2). Tmax data are more scattered than HI data. A similar trend is also observed on the PI vs. depth plot (not shown), the PI varying from 5% in Well D (early mature) to 30-50% in well A (late mature). Assuming a similar initial kerogen with an initial HI=550 mg/g (conservative hypothesis), then the Transformation Ratio (TR) can be assessed: in Well D, Tuwaiq-SR HI=530 mg/g TR=6% at 6228 ft MD (1835 m TVDrkb); in Well B, Tuwaiq-SR HI=350 mg/g TR=51% at 7789 ft MD (2374 m TVDrkb); in Well A, Tuwaiq-SR HI=180 mg/g TR=79% at 9580 ft MD (2920 m TVDrkb).

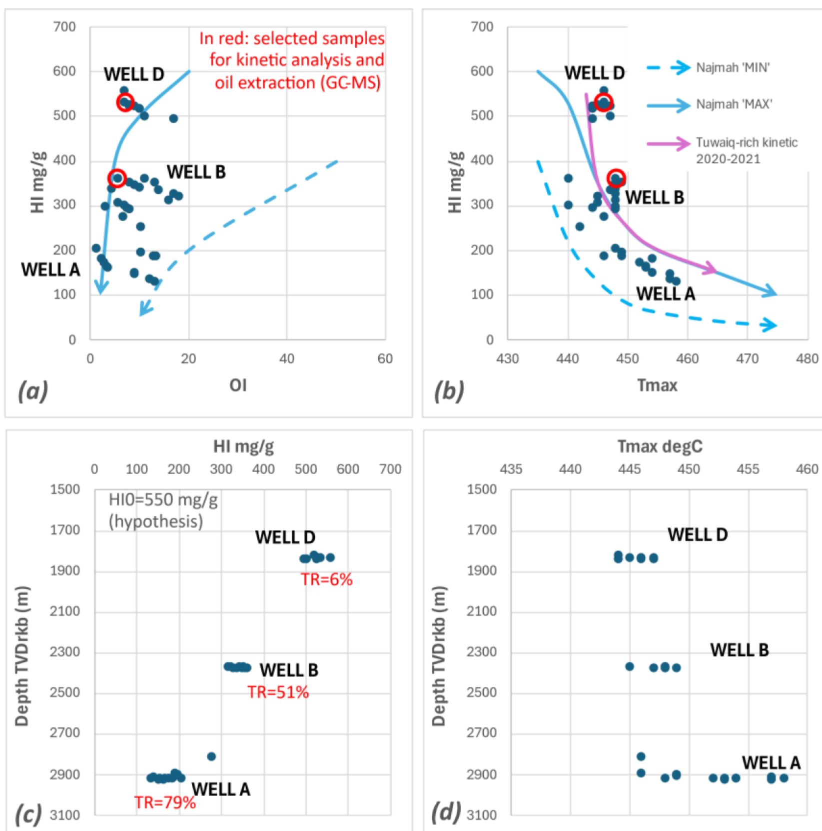


Figure 2—Classical plots displaying Rock-Eval® data for Well A-B-D (samples with OI < 20 mg/g, data from IFPEN laboratories in 2020 and other third-party laboratories in 2022): (a) HI/OI; (b) HI/Tmax; (c) Depth/HI; (c) Depth/Tmax.

Kinetic Analysis

It is possible to compare kinetic schemes with Rock-Eval® data using a kinetic simulator (such as those included in petroleum system modeling tools). It becomes immediately apparent that the kinetic schemes derived from the Well B (more mature) and Well D (less mature) samples are superimposable – indicating similar behavior – though they originate from different starting points on the HI vs. Tmax plot due to differing maturity levels. This confirms a high degree of similarity between the two analyzed kerogens. This kerogen is certainly marine Type II, but its maturity trend shows affinities with Type I, a behavior characteristic of some carbonate-rich, sulfur-rich kerogens – especially those found in the Gulf region (such as the Najmah kerogen in Kuwait). Both kinetic models also show good agreement with the Rock-Eval® data (Tmax/HI), particularly for samples selected with high OI. The scatter observed around this calibration reflects both measurement uncertainties and natural variability in the kerogen within the Tuwaiq-SR (probable oxidation of the organic matter).

In addition, bulk kinetics can be used to estimate a more precise level of thermal maturity. Elements of this workflow, previously presented in conferences (e.g., Dubille et al., 2022), involve analyzing the transformation ratio (TR) associated with each activation energy, considered independently at a given maturity level, and the shape of the activation energy spectrum (Figure 3). This method is particularly effective when at least two samples of the same kerogen are available at different maturity levels, as is the case in this study.

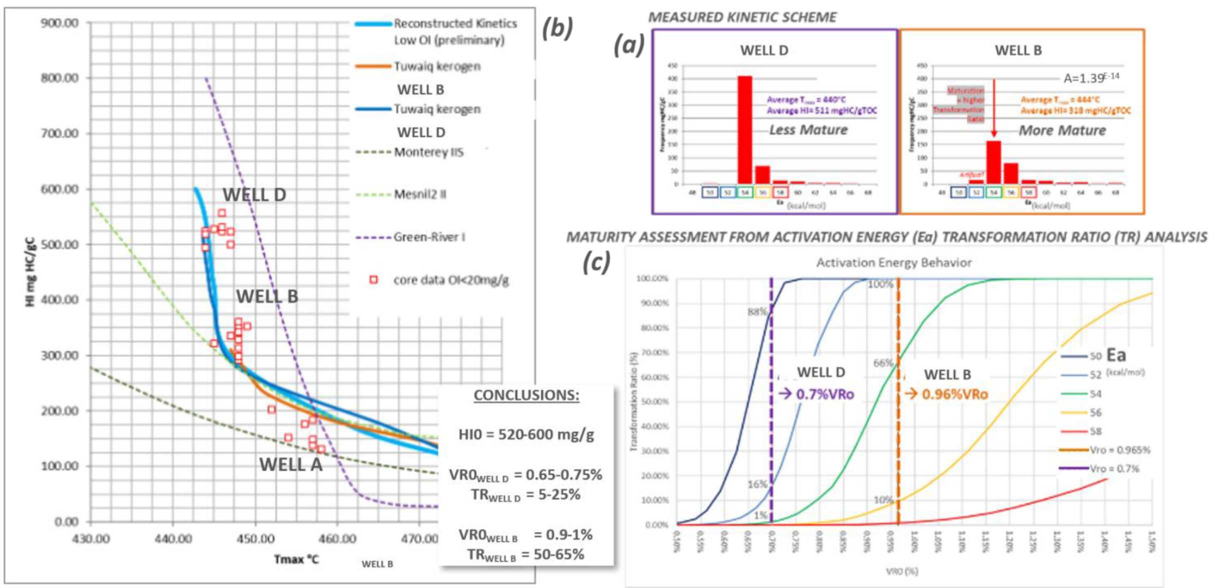


Figure 3—Bulk kinetic scheme analysis (open system): (a) Kinetic schemes (activation energy spectrum) for two organic-rich samples from Well B & Well D; (b) Kinetic scheme modelling and plot with Rock-Eval® data (HI/Tmax); (c) Maturity level assessment from activation energy transformation ratio analysis.

We obtain the following results: in Well D Tuwaiq Mountain maturity level = 0.65-0.7%VRoEq at 6228 ft MD (1837 m TVDrkb); in Well B, Tuwaiq Mountain maturity level = 0.9-1%VRoEq at 7789 ft MD (2374 m TVDrkb). The two wells are located close to each other, the main difference being the burial of the source rock. Such a pronounced maturity contrast (>0.2%) over a limited vertical distance (476 m) is a noteworthy finding.

Organic Petrography

Based on observations of organic petrography and palynofacies, all 50 of the investigated rock samples show the presence of solid bitumen and amorphous organic matter, predominantly aquatic organic matter that is oil-prone. This is consistent with the Rock-Eval® data. As vitrinite was scarce or absent, reflectance was measured exclusively on solid bitumen, and then the equivalent vitrinite reflectance VRoeq was calculated using the formula proposed by Schoenherr et al. (2007): %VRoeq = (%BRo + 0.2443)/1.0495, or alternatively the proposed formula by Jacob (1989): %VRoeq = 0.618 * %BRo + 0.4 (the result is quite different at high maturity level, for example a BRo = 1.13% gives 1.31% VRoeq with Schoenherr's formula and 1.1% with Jacob's formula). Using solid bitumen reflectance and a conversion formula necessarily introduces higher uncertainty and imprecision into the evaluation of the thermal maturity level. In this instance, it is recommended that samples with a higher TOC content and apparent reflectance are used as a reference, although residual anomalies cannot be ruled out.

Analyzed Jurassic samples (Figure 4) range from the early oil window to the late oil/condensate window: 0.65 to 0.7% VRoeq at ~6000 ft. deep TVD in Well D well, 0.8 to 1% VRoeq at ~7800 ft deep TVD in Well B (lower Tuwaiq Mountain organic-rich layer), 1.1 to 1.3% VRoeq (depending on the conversion law) at ~9500 ft deep TVD in Well A where Tuwaiq-SR is reaching the end of the hydrocarbon generation window as also indicated by the Rock-Eval® data (Tmax and HI). At this stage, it can be said that the thermal maturity level is quite high compared to the depth (Between the Khaleej Al Bahrain Basin and the Awali dome, the top of the oil window is at about 5000-6000 ft (1600-1800 m) deep, the top of the Wet Gas window is at about 9000-11000 ft (2800-3200 m) deep.

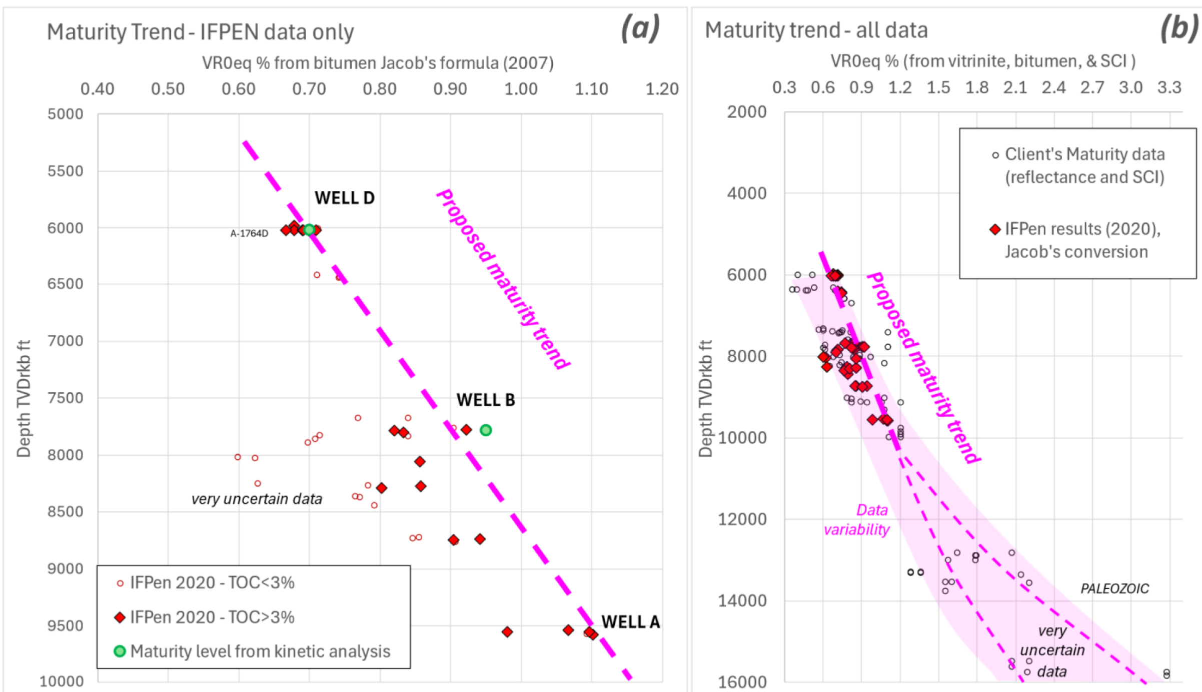


Figure 4—Maturity trends from Bitumen Reflectance measurements BRo (Depth/VRoeq, VRoeq computed with Jacob's formula): (a) Zoom on Jurassic samples analyzed by the IFPEN, the green dots correspond to the estimate of the "absolute" maturity level based on the two kinetic analyses on kerogen samples from wells D and B; (b) Same plot with all available samples, including maturity indicators from Paleozoic units (very uncertain and poorly constrained data from third-party laboratories, estimated from bitumen reflectance and SCI).

Fluid Inclusions

In the calcite vein sampled in Well B (Hanifa Formation, 7672 ft MD, 2338 m TVDrkb), two-phase secondary fluid inclusions (water and gas) with a homogenization temperature between 120 and 135 °C (isochore heating of fluid inclusions) could be identified (Figure 5). Homogenization temperature refers to the temperature at which the two-phase (liquid–vapour) fluid in fluid inclusions becomes single-phase during heating. This is commonly used in fluid inclusion studies to estimate the minimum temperature at which the fluid was trapped in the inclusion (the actual trapping temperature could be higher). Additionally, the maximum thermal stress experienced by the rock may not be reflected in the observed fluid inclusions.

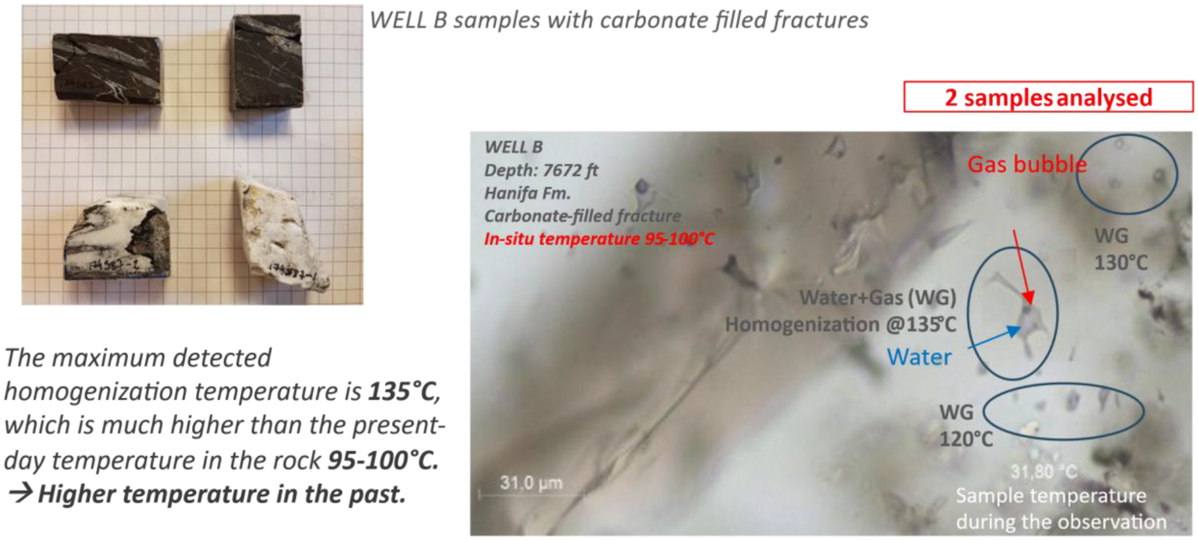


Figure 5—Fluid Inclusion Analysis (Well B samples).

According to the corrected BHT in neighboring wells, the present-day temperature is only 95–105 °C. This implies that temperatures were at least 15 °C higher in past geological conditions, possibly 20–40 °C higher. However, taken on its own, this result is not highly conclusive due to uncertainties related to the methodology. Furthermore, the fluid responsible for forming the sampled calcite vein could correspond to a locally circulating hydrothermal fluid within the rock, which may not be representative of the broader thermal field and petroleum system. However, this result is supported by observations from other maturity indicators.

Assessment of the Oil Maturity Level (Oil samples & Source Rock extracts)

Terpane analysis (GC-MS m/z 191) provides reliable maturity markers for characterizing oils and source rocks extracts in the Gulf region (cf. Baniasad et al., 2023). Combining all the data available (from IFPEN laboratories and from previous geochemical studies, Figure 6), a very clear maturity trend is emerging (despite the high uncertainty inherent in this empirical method). Considering the (Ts/Ts+Tm) ratio it appears that: the source rock extract from Well D Tuwaiq-SR is Early Mature (about 0.7 to 0.8% VRoeq); the source rock extracts and oil samples from Well B is Mid to Late Mature (about 0.9 to 1.1% VRoeq); the source rock extracts and oil samples from Well A in the Khaleej Al Bahrain is Late Mature (about 1.1 to 1.3% VRoeq). These findings are fully consistent with previous results. Plotting the apparent maturity level of the oils vs. depth, it is possible to assess the regional maturity trend, considering that the maturity of the oil cannot be higher than the maturity of the "host rock" (reservoir rock or source rock) due to the secondary cracking. This maturity level based on biomarker data is coherent with the maturity level based on the analysis of rock samples (BRo, Rock-Eval® and Kinetic analyses, etc.).

Note that the analysis of the oil composition and properties also yielded interesting results: the Saturate / (Saturate + Aromatic + Resin) ratio appears to correlate well with the maturity level of the oils, as estimated from biomarkers. A correlation is also observed with API gravity, though it is less consistent due to greater scatter and outliers. These findings provide additional support for the interpretation of oil maturity, and maturity levels.

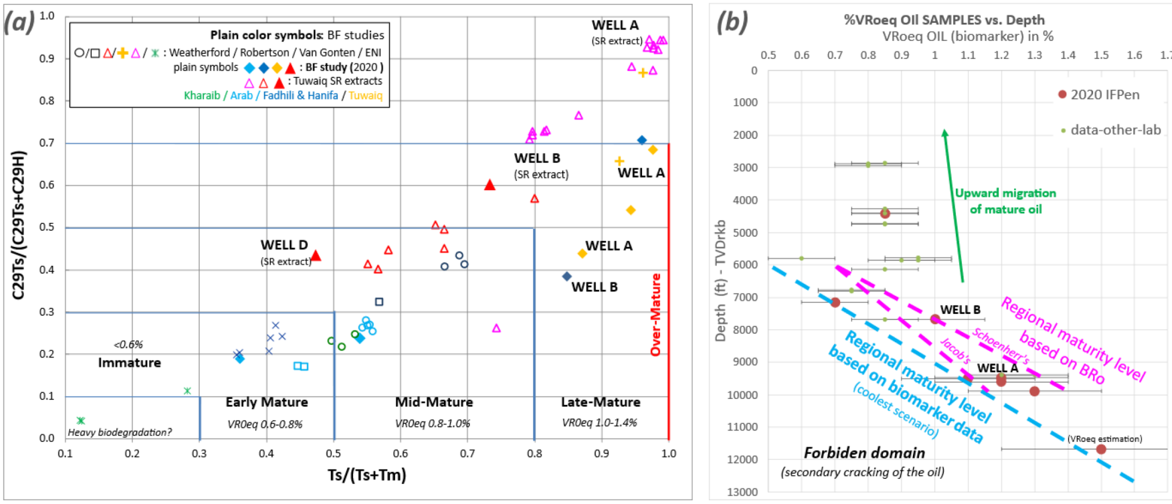


Figure 6—Maturity level from biomarker analysis (GC-MS m/z 191, Terpanes): (a) C29Ts/(C29Ts+C29H) vs. Ts/(Ts+Tm), oil samples and source rock extracts (all available data), VRoeq assessment; (b) Depth/VRoeq estimated for oil samples (all available data).

Discussion on the Origin of the Thermal Event

Synthesis of the Observations

Present-day corrected temperature measurements (BHT and MDT) indicate that the basin has a thermal gradient of 28 to 34 °C/km (computed from the surface; the average surface temperature is currently around 26 °C onshore) and an average of 30 °C/km at great depths. In fact, the thermal gradient remains relatively constant throughout the study area, except in the northern offshore domain, where it is slightly lower. The temperature data primarily comes from well tests and is reliable; it cannot be significantly underestimated. Given the age of the analyzed sediments (Mesozoic) and the structuring of the basin, a preliminary assessment of maturity levels based on present-day temperatures suggests that the entrance to the oil window (0.6% VRo) should be at around 2,400-2,500 m (~7900-8200 ft), while the entrance to the wet gas window (1.2% VRo) should be at around 4,000-4,500 m (13100-14700ft).

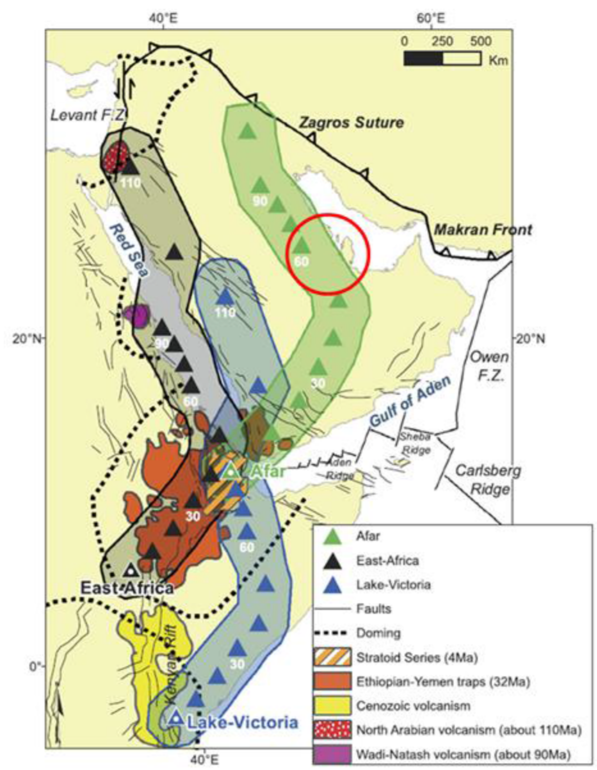
On the other hand, all maturity markers indicate that the rocks and oils have undergone significant thermal stress which increases rapidly with depth (except in northern wells). Despite multiple uncertainties, the convergence of all the data rather suggests that the entrance within the oil window should be around 1600-2000 m (present day temperature 75-85 °C +-10%), while the entrance within the wet gas window should be around 2700-3200 m (present day temperature 110-120 °C +-10%). This implies that the maximum temperature in the past was 20 to 30°C higher than the current temperature, or even more, which seems to be corroborated by the homogenization temperature observed in some fluid inclusions (samples from Well B).

Therefore, it is necessary to introduce a "thermal event" into the basin's past geological history. This event could be one of two things: either the higher temperature in the past is simply explained by greater burial, which necessarily implies significant intermediate erosion, or the geothermal gradient was higher in the past, implying the existence of a transient phenomenon that affected the lithosphere under Bahrain in the past. A combination of erosion and lithospheric activity is also possible. It should be noted that, while both conceptual models could explain the observed maturity levels in Jurassic sediments, the consequences for the maturity levels in deep sedimentary units would be very different. In the case of a higher thermal gradient, a steeper maturity trend would be expected in Mesozoic units, and a much higher maturity level would be expected in Paleozoic units. This appears to be the case, although the poor quality of the available data in Paleozoic layers makes it impossible to be certain. Further analysis requires the use of a basin model.

Basin Modelling for Testing Geological Scenarios

The uplift and erosion event that occurred during the Oligocene and Miocene periods (after 36 Ma) could be a candidate. The maximum burial (and thus temperature) may have occurred shortly after the Dammam Formation was deposited. However, according to our current geological knowledge, the erosion is located at the top of the Awali structure (Figure 1). In the reference geological scenario, the estimated eroded thickness is only 100 m, which is insufficient to explain the observed thermal maturity. In fact, an erosion exceeding 1000 m (requiring a "hot" lithosphere to explain the steep thermal gradient) would be needed to explain the observed maturity level, which seems out of reach. It is also difficult to invoke an older erosion event, as this would require more erosion. Nevertheless, this scenario has been tested using a basin model.

The other scenario implies strong heating of the lithosphere. A significant thermal anomaly of this kind in the past could be related to deep thermal mantle activity, as described by Vicente de Gouveia et al. (Figure 7). This activity may be associated with the migration of the Afar hotspot, which crossed the Arabian Peninsula from the Upper Cretaceous to the Paleogene period (60-50 Ma). High thermal gradients probably imply deep magmatism at the base of the crust (underplating). While this scenario is possible despite the lack of volcanic activity and massive thermal uplift, it has yet to be proven. A modern analogue could be the sedimentary basins of southern Algeria where thermal gradients >40°C/km are currently observed. This scenario has also been tested using a basin model.



The authors describe higher maturity levels along the trace of the Afar hotspot across Saudi Arabia, however clues are limited, more data is required. In the other hand the geodynamic reconstruction is convincing.

(From Vicente de Gouveia et al., 2018.)

Figure 7—Migration of deep mantellic hot spots in Africa and Arabic Peninsula since 110 Ma (from Vicente de Gouveia et al., 2018).

As a starting point, three alternative 1D basin models corresponding to Well B have been tested (Figure 8): 1) An initial model without a past thermal event, as a reference; 2) A scenario involving massive post-Dammam erosion; 3) A scenario involving a thermal event within the lithosphere during the Paleogene. Preliminary 1D basin modelling confirms that the observed maturity level cannot be reached at present-day temperatures; temperatures were much higher in the past. Scenario 3, which involves a past thermal event (Paleogene), is the most efficient and the most likely, even though the origin of the thermal event remains uncertain. It has therefore been selected for further analysis with the 3D model.

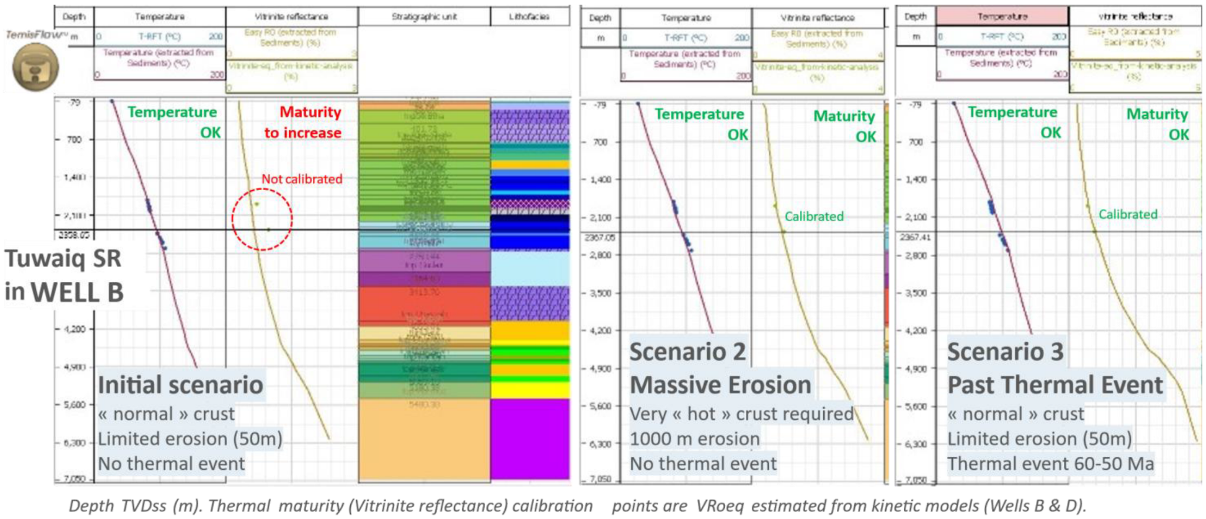


Figure 8—TemisFlow® 1D basin model testing alternative scenarios explaining high maturity levels in Bahrain. There are 3 scenarios corresponding to the Well B location, with temperature and thermal maturity (VRoeq) calibration. Scenario 3 seems more likely.

To calibrate the present-day temperature, a lithospheric model with the following characteristics was applied: upper crust: 25-30 km thick with radiogenic heat production of about 5.0E^{-6} W/m^3; lower crust: 11 km thick with low radiogenic heat production of 4.0E^{-7} W/m^3; upper mantle: 50 km thick with no radiogenic heat production The simulated present-day heat flow at the top of the basement (at the base of the sediments) varies from 60 to 67 mW/m^2, decreasing towards the northeast. To calibrate past temperatures and the observed thermal maturity of Tuwaiq-SR samples, a thermal event is added to the lithosphere during the Paleogene period (68–49 Ma). This induces a lateral variation in heat flow, decreasing towards the northeast and increasing towards the southwest (Figure 9). This model provides the best match with well calibration data, including data from deep Paleozoic units (Figure 10).

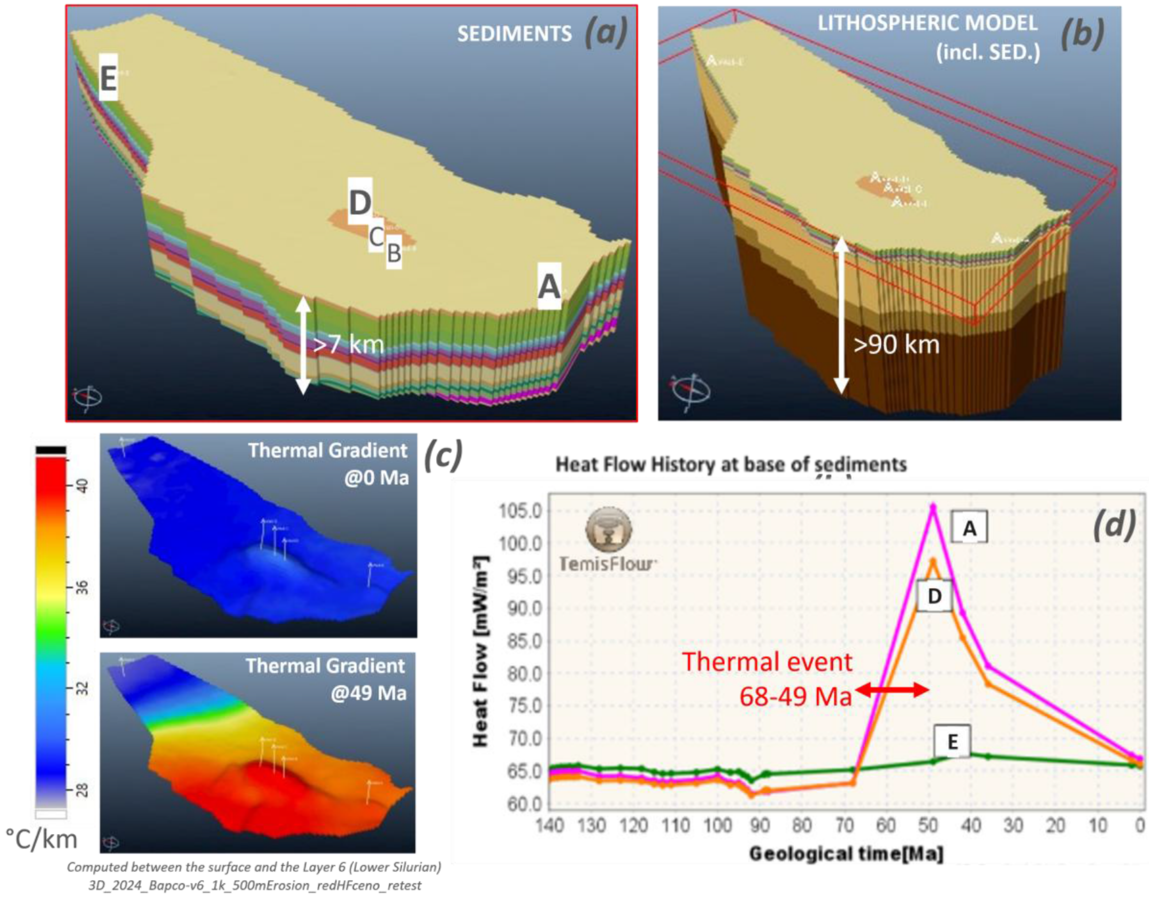


Figure 9—TemisFlow® 3D basin modelling results: (a) & (b) Overview of the 3D geological model (sedimentary cover and lithosphere); (c) Average thermal gradient computed at present day and 49Ma ago at the end of the thermal event; (c) Modelized evolution of the basal Heat Flow at the base of the sediment, at 3 locations (Well A, Well D, Well E, modelling result).

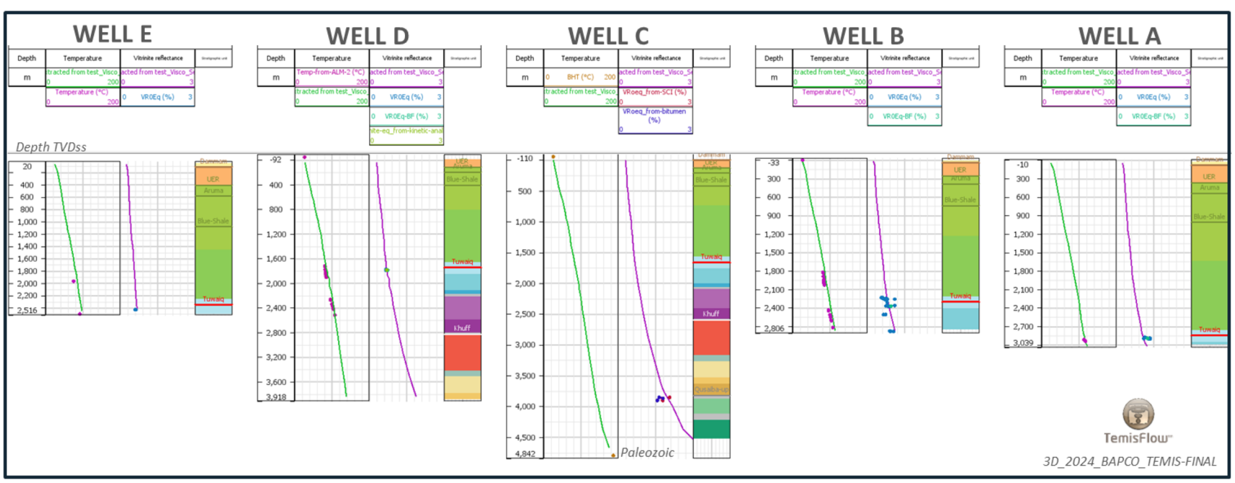


Figure 10—TemisFlow® 3D "calibration plots" (temperature and VRoeq) for Wells A/B/C/D. Calibration point for "Vitrinite reflectance" curves have different origins: Light blue points = VRoeq from BRo (conversion with Schoenherr's formula); Light green points = interpretation of kinetic models (Wells D & B); Purple and red points (Well C) = data from third-party laboratories, estimated from bitumen reflectance and SCI.

Conclusions

This study demonstrates the value of combining various maturity markers and incorporating the outcomes into a basin model. One of the project's most striking findings is the identification of a significant past thermal event. This important element for understanding the petroleum system was not mentioned in previous studies. The cause remains unclear, but it is certainly related to a lithospheric process of relatively large scale (>75 km radius). It is unlikely that the observed maturity anomalies are solely the result of locally occurring hydrothermal fluid flow, even if such fluids could have existed and accentuated the phenomenon. It is also possible that the thermal event occurred alongside erosion of relatively small magnitude. These hypotheses require confirmation.

If verified, the implications of this thermal event for exploring unconventional plays in Bahrain and Saudi Arabia would be significant, as it could change our understanding of the thermal history of other supposedly "stable" sedimentary basins. Notably, thermal anomalies have also been observed further west in Saudi Arabia, in the Jafurah basin. Hakami and Inan (2016) required a higher thermal gradient to calibrate a well in the northern part of their study area, i.e. towards the Bahraini border. Unfortunately, only converted Tmax is used as a maturity indicator (Tmax up to 490°C in Jurassic source rocks), and the anonymized data is difficult to reuse. Conversely, no such thermal anomaly has been described further east (in Qatar).

In the current state of knowledge, we do not have an irrefutable explanation for our observations. Further investigation is needed to clarify this thermal anomaly. For example, thermochronometry could tentatively be used to date the thermal event (e.g. zircon and/or apatite fission track analysis on clastic sediments from Tertiary, Triassic and Paleozoic units). Additional fluid inclusion analysis coupled with carbonate phase dating would also be valuable. More advanced sensitivity and uncertainty analyses will be performed.

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